Vishaal Kumar

vishaalk@seas.upenn.edu | 267.637.6177
linkedin.com/in/vishaalpkumar | github.com/VishaalPKumar | vishaalpkumar.github.io

EDUCATION

UNIVERSITY OF PENNSYLVANIA
BSE in Computer Science
May 2023 | Philadelphia, PA
School of Engineering
GPA: 4.0 / 4.0

COURSEWORK

COMPLETED
- Mathematical Foundations of Computer Science
- Programming Languages and Techniques
- Multi-variable Calculus
- Market and Social Systems on the Internet

CURRENT
(Fall 2019)
- Data Structures and Algorithms
- Scalable and Cloud Computing
- Linear Algebra
- Probability

(Tentative - Spring 2021)
- Introduction to Computer Architecture

STRENGHTS

LANGUAGES
• Java • Swift • Python • OCamL

FRAMEWORKS
• Google Firebase • CoreML2
• NumPy • Selenium • Tensorflow
• Keras

OTHER POSITIONS

CTSA ANXIETY APP
(Recently Selected)
Software Developer
July 2020 - Present

DAILY PENNSYLVANIAN
Product Analyst
Jan 2020 - Present

HATCH
(An early venture startup)
Technical Fellow
Jan 2020 - May 2020

MCAFEE
Software Engineering Intern (Summer 2020)
Cancelled due to COVID-19

WORK EXPERIENCE

GALLOP WEALTH | Innovations Team Intern
May 2020 - Aug 2020 | , Remote/Held Virtually
- Devised rating algorithms for calculating the performance of stock portfolios, such as the Sharpe Ratio, using Pandas and NumPy in Python
- Created a model to assign a rating (out of 5) for various aspects of a stock portfolio. The model adjusts itself with the changes in the structure of data.
- Spearheaded the development for the twitter stock analyzer for the main mobile application. Using Xcode's CoreML2 and NLP, the bot scrapes the latest relevant Twitter posts and uses sentiment analyst to help predict stock movements.

UPENN COMPUTER SCIENCE SOCIETY | Vice President
September 2019 – Present | Philadelphia, PA
- Responsibilities include establishing a general vision, leading the executive board, hosting events, and maintaining strong relations with the CIS department.

UNIVERSITY OF PENNSYLVANIA | Tutor
Jan 2020 – Present | Philadelphia, PA
- Tutor for the main introductory to programming course (CIS 110) and for Calculus 2 (MATH 104).
- Responsible for working with individuals or student groups to clarify course content and promote academic independence.

DR. CHAMITH S. RAJAPAKSE, PH.D. | Research Assistant
May 2020 - Present | Philadelphia, PA
- Assisting with research centered around the use of Artificial Intelligence in automated image segmentation in Radiology
- Conducted training of data and studied the use of Tensorflow and Keras in machine learning in regard to medical imaging, specifically in the classification of X-ray scans.

TECH IT OUT PHILLY | Student Mentor
Jan 2020 – Present | Philadelphia, PA
- Mentor in a program to teach low-resource Philly high school students computer science through hands-on coding and hardware experiments.

PROJECTS

BOOKSTAGRAM | iOS Application
Swift | Google Firebase, Xcode
Developed a clone of Instagram, specific to the sharing of book reviews and ratings. Built in Xcode and utilizes the Google Firebase framework in order to enable cloud storage, user authentication and real-time database management.

PAC-VIRUS | 2-D Game
Java | Eclipse
Developed a custom version of Pac-Man using the Java Swing GUI framework that commemorates the work of health workers during the COVID19 pandemic. Utilized inheritance, complex data structures and created an AI for the movement of enemy sprites.

AMAZON PRICE TRACKER | Automated Interface
Python | PyCharm
Developed an automated program using Selenium that scrapes data from Amazon in order to notify users about price changes in interested Amazon products. The program updates a Google Spreadsheet and sends relevant emails to the user.

CHECK IT! | iOS Application
Swift | Google Firestore, Swift
Developed an iOS Application that allows users to track daily tasks by sorting them into categories. Users are authenticated using Firebase and all user data is stored in a NoSQL database (Firestore)